

Penalized and Debiased GMM with Potentially Invalid Moment Conditions

Tu Li

Department of Economics
University of Arizona

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Moment selection with an estimated first step

- ▶ Valid baseline moments identify the parameter of interest.
- ▶ Additional moments can improve precision, but some may be invalid.
- ▶ Using only baseline moments can discard information. Imposing invalid candidate restrictions can cause inconsistency.
- ▶ Nonparametric or machine-learning first steps introduce estimation error that also affects moment selection.

Contribution

- ▶ A penalized debiased GMM procedure jointly estimates the parameter and selects valid candidate moments.
- ▶ Orthogonalization and cross-fitting account for estimated nuisance functions.
- ▶ Under the stated conditions, selection is consistent and the estimator has the oracle debiased GMM distribution.
- ▶ Simulations support an unpenalized post-selection refit (Post-PDG) for final estimation and inference.

The theory maintains a fixed number of moments and a fixed DGP.

Related literature

- ▶ **GMM moment selection**

Andrews (1999); Andrews and Lu (2001); Hong et al. (2003).
Criteria for choosing moment restrictions.

- ▶ **Penalized moment selection**

Liao (2013); Cheng and Liao (2015); Caner et al. (2018); Chang et al. (2023). Auxiliary means encode candidate invalidity.

- ▶ **Semiparametric GMM and orthogonal moments**

Newey (1994); Akerberg et al. (2014); Chernozhukov et al. (2018, 2022).

- ▶ **Selection with a nonparametric first step**

Ouyang and Yang (2025) study a binary-choice model with a particular nonparametric first-step setting. This paper develops a general semiparametric GMM framework.

Baseline and candidate moments

Let W have distribution F_0 , let θ_0 be the d_θ -dimensional target, and let γ_0 be the first-step function. All vectors are column vectors; tuples denote vertical stacking.

$$E[g_1(W, \gamma_0, \theta_0)] = 0,$$

$$E[g_2(W, \gamma_0, \theta_0)] = \mu_0.$$

The functions g_1 and g_2 have m_1 and m_2 components.

- ▶ The baseline moments uniquely identify θ_0 .
- ▶ Candidate j is valid when $\mu_{0,j} = 0$ and invalid when $\mu_{0,j} \neq 0$.
- ▶ The locations of the zero and nonzero components are unknown.

Augmented moments and the invalid set

Introduce an auxiliary mean μ_j for each candidate moment:

$$g^*(W, \gamma, \theta, \mu) = \begin{bmatrix} g_1(W, \gamma, \theta) \\ g_2(W, \gamma, \theta) - \mu \end{bmatrix}.$$

At the truth, $E[g^*(W, \gamma_0, \theta_0, \mu_0)] = 0$.

$$\mathcal{S} = \{j : \mu_{0,j} \neq 0\}.$$

- ▶ $\mathcal{S}$ is the **invalid set**.
- ▶ $\mathcal{S}^c$ is the set of **valid candidates** to retain.
- ▶ Selection becomes recovery of the support of μ_0 .

Accounting for the first step: the FSIF

To account for first-step estimation, compute the FSIF of g^* .

Let $\gamma(F)$ be the first-step probability limit under F , with $\gamma_0 = \gamma(F_0)$. For an alternative distribution H , set $F_t = (1 - t)F_0 + tH$, $t \in [0, 1]$.

For every admissible H and every $\theta \in \Theta$, holding μ fixed and the outer expectation under F_0 ,

$$\left. \frac{d}{dt} E[g^*(W, \gamma(F_t), \theta, \mu)] \right|_{t=0} = \int \phi(w, \gamma_0, \alpha_0, \theta) H(dw),$$

$$E[\phi(W, \gamma_0, \alpha_0, \theta)] = 0.$$

Here $\phi = (\phi_1, \phi_2)$ is the first-step influence function and α_0 is an additional nuisance function.

From the FSIF to the debiased moments

Because μ is fixed along the path F_t ,

$$\left. \frac{d}{dt} \mathbb{E}[g_2(W, \gamma(F_t), \theta) - \mu] \right|_{t=0} = \left. \frac{d}{dt} \mathbb{E}[g_2(W, \gamma(F_t), \theta)] \right|_{t=0}.$$

Thus, g^* and $g = (g_1, g_2)$ have the same FSIF $\phi = (\phi_1, \phi_2)$, with no μ argument.

Construct the debiased augmented moment function as

$$\psi^*(W, \gamma, \alpha, \theta, \mu) = g^*(W, \gamma, \theta, \mu) + \phi(W, \gamma, \alpha, \theta).$$

The auxiliary mean μ continues to account for candidate invalidity.

Local and global nuisance insensitivity

Under the FSIF regularity conditions, for every $\theta \in \Theta$:

Locally in γ : for any admissible direction δ ,

$$\left. \frac{d}{dt} \mathbb{E}[\psi^*(W, \gamma_0 + t\delta, \alpha_0, \theta, \mu)] \right|_{t=0} = 0.$$

Globally in α at γ_0 : for every $\alpha \in \mathcal{A}$, where $\alpha_0 \in \mathcal{A}$,

$$\mathbb{E}[\phi(W, \gamma_0, \alpha, \theta)] = 0,$$

and hence

$$\mathbb{E}[\psi^*(W, \gamma_0, \alpha, \theta, \mu)] = \mathbb{E}[g^*(W, \gamma_0, \theta, \mu)].$$

These are properties of the population moments (Chernozhukov et al., 2022).

Cross-fitted sample moments

Partition the i.i.d. sample into folds $I_1, \dots, I_L$. Estimate $\hat{\gamma}_\ell$, $\hat{\alpha}_\ell$, and $\tilde{\theta}_\ell$ using observations outside I_ℓ .

$$\bar{\psi}^*(\theta, \mu) = \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left[g^*(W_i, \hat{\gamma}_\ell, \theta, \mu) + \phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) \right].$$

- ▶ The FSIF remains fixed at $\tilde{\theta}_\ell$ during minimization.
- ▶ With orthogonality and suitable rate conditions, cross-fitting makes first-step effects negligible at the $\sqrt{n}$ scale.

The PDG estimator

$$(\hat{\theta}, \hat{\mu}) = \arg \min_{\theta, \mu} \left\{ \bar{\psi}^*(\theta, \mu)^T V_n^* \bar{\psi}^*(\theta, \mu) + \sum_{j=1}^{m_2} \hat{P}(\mu_j) \right\},$$

where V_n^* is positive definite.

- ▶ Strong shrinkage near zero selects candidate moments as valid.
- ▶ A small penalty slope away from zero limits bias on nonzero violations.

An example is the adaptive Lasso penalty:

$$\hat{P}(\mu_j) = \frac{\lambda_n}{|\tilde{\mu}_j|^r} |\mu_j|, \quad r > 0,$$

where $\tilde{\mu}_j$ is a preliminary estimate of $\mu_{0,j}$.

Example 1: Endogenous partially linear model

Following Robinson (1988), consider

$$Y = X\theta_0 + h_0(U) + \epsilon, \quad E[\epsilon | U] = 0.$$

The regressor X may be endogenous. Instruments satisfy

$$E[Z_1\epsilon] = 0, \quad E[Z_2\epsilon] = \mu_0.$$

- ▶ Z_1 is a valid baseline instrument.
- ▶ Z_2 is a candidate instrument with unknown validity.

Baseline identification requires

$$E[(Z_1 - E[Z_1 | U])(X - E[X | U])] \neq 0.$$

Example 1: Debiased moment conditions

The model implies

$$Y - E[Y | U] = \{X - E[X | U]\}\theta_0 + \epsilon.$$

The augmented debiased moment conditions are

$$E \left[\{Z_1 - E[Z_1 | U]\} (\{Y - E[Y | U]\} - \{X - E[X | U]\}\theta) \right] = 0,$$

$$E \left[\{Z_2 - E[Z_2 | U]\} (\{Y - E[Y | U]\} - \{X - E[X | U]\}\theta) \right] - \mu = 0.$$

The unknown functions $E[Y | U]$, $E[X | U]$, $E[Z_1 | U]$, and $E[Z_2 | U]$ must be estimated in the first step.

See Newey (1994) and Ichimura and Newey (2022).

Example 2: Endogenous single-index model

Following Birke et al. (2017),

$$Y = h_0(X^T \theta_0) + \epsilon, \quad E[\epsilon \mid Z_1] = 0.$$

Let Θ impose the usual scale and sign normalization. For every $\theta \in \Theta$, assume a unique profile NPIV solution:

$$E[Y - h_\theta(X^T \theta) \mid Z_1] = 0, \quad h_{\theta_0} = h_0.$$

- ▶ Z_1 supplies valid instruments for the first step.
- ▶ Z_2 supplies valid baseline moments that locally identify θ_0 .
- ▶ Z_3 supplies candidate moments and is excluded from the first step.

Example 2: Debiased moment conditions

Suppose square-integrable functions $\alpha_{q,\theta}$ exist with

$$E[\alpha_{q,\theta}(Z_1) \mid X^T \theta] = E[Z_q \mid X^T \theta], \quad q = 2, 3.$$

If $E[Z_2\epsilon] = 0$ and $E[Z_3\epsilon] = \mu_0$, the augmented debiased moments are

$$\begin{aligned} E\left[(Z_2 - \alpha_{2,\theta}(Z_1))(Y - h_\theta(X^T \theta))\right] &= 0, \\ E\left[(Z_3 - \alpha_{3,\theta}(Z_1))(Y - h_\theta(X^T \theta))\right] - \mu &= 0. \end{aligned}$$

The adjustment accounts for estimation of the profile function h_θ (Ichimura and Newey, 2022).

Assumption 1: Identification and first step

Write $g = (g_1, g_2)$ and $\eta = (\theta, \mu)$.

C1. Θ is compact and $E[g(W, \gamma_0, \theta)]$ is continuous in $\theta \in \Theta$.

Moreover,

$$E[g_1(W, \gamma_0, \theta)] = 0 \iff \theta = \theta_0.$$

C2. $V_n^* \xrightarrow{P} V^*$, where V^* is positive definite.

C3. For every $\ell = 1, \dots, L$,

$$\|\hat{\gamma}_\ell - \gamma_0\| = o_P(1).$$

Assumption 1: Moment continuity

C4. For each $\ell = 1, \dots, L$ and every $\theta \in \Theta$,

$$\int \|g(w, \hat{\gamma}_\ell, \theta) - g(w, \gamma_0, \theta)\| F_0(dw) = o_P(1),$$

and $E[\|g(W, \gamma_0, \theta)\|] < \infty$ for all $\theta \in \Theta$.

C5. There exist $C > 0$ and a nonnegative function $d(W, \gamma)$ such that, for $\|\gamma - \gamma_0\|$ sufficiently small and all $\theta_1, \theta_2 \in \Theta$,

$$\|g(W, \gamma, \theta_2) - g(W, \gamma, \theta_1)\| \leq d(W, \gamma) \|\theta_2 - \theta_1\|^{1/C},$$

where $E[d(W, \gamma)] < C$ for all such γ .

Assumption 1: The first-step correction

Let $\phi = (\phi_1, \phi_2)$.

C6. For every $\ell = 1, \dots, L$,

$$\int \|\phi(w, \hat{\gamma}_\ell, \alpha_0, \theta_0) - \phi(w, \gamma_0, \alpha_0, \theta_0)\|^2 F_0(dw) = o_P(1),$$

$$\int \|\phi(w, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(w, \gamma_0, \alpha_0, \theta_0)\|^2 F_0(dw) = o_P(1),$$

and

$$E[\|\phi(W, \gamma_0, \alpha_0, \theta_0)\|] < \infty.$$

The first condition controls the fitted $\hat{\gamma}_\ell$; the second controls $\hat{\alpha}_\ell$ and $\tilde{\theta}_\ell$.

Assumption 1: Interaction and penalty

C7. For every $\ell = 1, \dots, L$, define

$$\begin{aligned}\Delta_\ell(w) = & \phi(w, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) - \phi(w, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) \\ & - \phi(w, \hat{\gamma}_\ell, \alpha_0, \theta_0) + \phi(w, \gamma_0, \alpha_0, \theta_0).\end{aligned}$$

Then $\int \|\Delta_\ell(w)\| F_0(dw) = o_P(1)$.

C8. For each $j = 1, \dots, m_2$,

$$\hat{P}(\mu_j) \geq 0 \quad \text{for all } \mu_j \in \mathbb{R}, \quad \hat{P}(\mu_{0,j}) = o_P(1).$$

Theorem 1: consistency

Theorem 1

Let $\hat{\eta} = (\hat{\theta}, \hat{\mu})$ and $\eta_0 = (\theta_0, \mu_0)$. Under C1–C8,

$$\|\hat{\eta} - \eta_0\| = o_P(1).$$

- ▶ Under C1 and C3–C7, the sample-moment remainder converges uniformly.
- ▶ The auxiliary parameter cancels from the remainder

$$R_n(\theta) = \bar{\psi}^*(\eta) - E[g^*(W, \gamma_0, \eta)].$$

- ▶ Baseline identification and a negligible penalty at the truth then deliver joint consistency.

Assumption 2: Identification and moments

R1. For $q = 1, 2$, $E[g_q(W, \gamma_0, \theta)]$ is continuously differentiable near θ_0 , and

$$\frac{\partial}{\partial \theta^T} E[g_1(W, \gamma_0, \theta_0)]$$

has full column rank.

R2. For each $\ell = 1, \dots, L$,

$$\int \|g(w, \hat{\gamma}_\ell, \theta_0) - g(w, \gamma_0, \theta_0)\|^2 F_0(dw) = o_P(1).$$

Also,

$$\begin{aligned} E[\|g(W, \gamma_0, \theta_0)\|^2] &< \infty, \\ E[\|\phi(W, \gamma_0, \alpha_0, \theta_0)\|^2] &< \infty. \end{aligned}$$

Assumption 2: Interaction and population bias

The following conditions hold for each $\ell = 1, \dots, L$.

R3. The interaction remainder satisfies

$$\int \Delta_\ell(w) F_0(dw) = o_P(n^{-1/2}), \quad \int \|\Delta_\ell(w)\|^2 F_0(dw) = o_P(1).$$

R4. The correction satisfies

$$P\left(\int \phi(w, \gamma_0, \hat{\alpha}_\ell, \tilde{\theta}_\ell) F_0(dw) = 0\right) \longrightarrow 1.$$

R5. The recentered population remainder satisfies

$$\mathbb{E} [g(W, \hat{\gamma}_\ell, \theta_0) + \phi(W, \hat{\gamma}_\ell, \alpha_0, \theta_0)] - \begin{bmatrix} 0 \\ \mu_0 \end{bmatrix} = o_P(n^{-1/2}).$$

Assumption 2 (R6): Local smoothness

For $\|\gamma - \gamma_0\|$ sufficiently small, $g(W, \gamma, \theta)$ is continuously differentiable in θ on a neighborhood $\mathcal{N}$ of θ_0 .

There exist $C > 0$ and a nonnegative function $d(W, \gamma)$ such that, for all $\theta \in \mathcal{N}$,

$$\left\| \frac{\partial}{\partial \theta^T} g(W, \gamma, \theta) - \frac{\partial}{\partial \theta^T} g(W, \gamma, \theta_0) \right\| \leq d(W, \gamma) \|\theta - \theta_0\|^{1/C},$$

where $E[d(W, \gamma)] < C$ for all such γ .

Assumption 2 (R6): Derivative convergence

The Jacobian at the truth is integrable:

$$\mathbb{E} \left[\left\| \frac{\partial}{\partial \theta^T} g(W, \gamma_0, \theta_0) \right\| \right] < \infty.$$

Differentiation can be interchanged with expectation at θ_0 .

For each $\ell = 1, \dots, L$,

$$\int \left\| \frac{\partial}{\partial \theta^T} g(w, \hat{\gamma}_\ell, \theta_0) - \frac{\partial}{\partial \theta^T} g(w, \gamma_0, \theta_0) \right\| F_0(dw) = o_P(1).$$

R6 controls the sample Jacobian and changes in the local stochastic remainder.

Assumption 2 (R7): Penalty regularity

The penalty satisfies $\hat{P}(0) = 0$. Let $\mathcal{S} = \{j : \mu_{0,j} \neq 0\}$ be the invalid support.

For each $j \in \mathcal{S}$, there is a fixed neighborhood $\mathcal{N}_j$ of $\mu_{0,j}$ such that

- ▶ $0 \notin \mathcal{N}_j$;
- ▶ $\hat{P}$ is twice continuously differentiable on $\mathcal{N}_j$ with probability approaching one;
- ▶ the curvature is locally uniformly negligible:

$$\sup_{\mu \in \mathcal{N}_j} |\hat{P}''(\mu)| = o_P(1).$$

Uniform control over $\mathcal{N}_j$ also controls the penalty curvature at intermediate points of an expansion.

Why the rate conditions matter

Under C6 and R2–R5,

$$\sqrt{n}\bar{\psi}^*(\eta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[g^*(W_i, \gamma_0, \theta_0, \mu_0) + \phi(W_i, \gamma_0, \alpha_0, \theta_0) \right] + o_P(1).$$

- ▶ Cross-fitting and mean-square convergence control the centered empirical error.
- ▶ R3–R5 control interaction terms and population bias.
- ▶ Feasible moments have the same first-order behavior as moments evaluated at the true nuisance functions.

Theorem 2: rate of convergence

Theorem 2

Under C1–C8 and R1–R7,

$$\|\hat{\eta} - \eta_0\| = O_P\left(\max\{n^{-1/2}, r_n\}\right),$$

where

$$r_n = \max_{j \in \mathcal{S}} |\hat{P}'(\mu_{0,j})|.$$

- ▶ $n^{-1/2}$ is the stochastic estimation error; r_n measures shrinkage on the nonzero violations.
- ▶ If $r_n = o_P(n^{-1/2})$, the estimator is root- n consistent. Exact selection also requires strong shrinkage near zero.

Assumption 3: Selection consistency

S1. The penalty derivative on the invalid support satisfies

$$\max_{j \in \mathcal{S}} |\hat{P}'(\mu_{0,j})| = o_P(n^{-1/2}).$$

S2. For any $j \in \mathcal{S}^c$ and any random sequence $\{\mu_{j,n}\}_n$ with $\mu_{j,n} = O_P(n^{-1/2})$, there exists a positive sequence a_n such that

$$\frac{a_n \lambda_n}{|\hat{P}'(\mu_{j,n})|} 1_{\{\mu_{j,n} \neq 0\}} = O_P(1), \quad \sqrt{n} a_n \lambda_n \longrightarrow \infty.$$

The penalty is asymptotically negligible at nonzero violations, but dominates sampling error near zero.

Theorem 3: Oracle Properties

Let $\hat{\mathcal{S}}$ and $\mathcal{S}$ denote the support of $\hat{\mu}$ and μ_0 , respectively, and define $\hat{\eta}_{\mathcal{S}} := (\hat{\theta}, \hat{\mu}_{\mathcal{S}})$ and $\eta_{0,\mathcal{S}} := (\theta_0, \mu_{0,\mathcal{S}})$. Under **Assumptions 1, 2, and 3**, we have

$$\lim_{n \rightarrow \infty} P(\hat{\mathcal{S}} = \mathcal{S}) = 1.$$

In addition, define

$$G^* := \frac{\partial}{\partial \eta_{\mathcal{S}}^T} E[g^*(W, \gamma_0, \eta_0)].$$

Then

$$\begin{aligned} \sqrt{n}(\hat{\eta}_{\mathcal{S}} - \eta_{0,\mathcal{S}}) = & - [(G^*)^T V^* G^*]^{-1} (G^*)^T V^* \\ & \times \left[\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi^*(W_i, \gamma_0, \alpha_0, \theta_0, \mu_0) \right] + o_P(1). \end{aligned}$$

The oracle GMM benchmark

Retain the baseline and valid candidate moments and their FSIF:

$$g_o(W, \gamma, \theta) = \begin{bmatrix} g_1(W, \gamma, \theta) \\ g_{2, S^c}(W, \gamma, \theta) \end{bmatrix},$$

$$\phi_o(W, \gamma, \alpha, \theta) = \begin{bmatrix} \phi_1(W, \gamma, \alpha, \theta) \\ \phi_{2, S^c}(W, \gamma, \alpha, \theta) \end{bmatrix}.$$

Because g_o contains g_1 , θ_0 uniquely solves $E[g_o(W, \gamma_0, \theta)] = 0$.
Define the original-moment Jacobian

$$G = \frac{\partial}{\partial \theta^T} E[g_o(W, \gamma_0, \theta_0)],$$

Oracle covariance and optimal weighting

The debiased-moment covariance is

$$\Omega = E[(g_o + \phi_o)(g_o + \phi_o)^T],$$

with $g_o(W, \gamma_0, \theta_0)$ and $\phi_o(W, \gamma_0, \alpha_0, \theta_0)$ evaluated at the truth.

With a consistent first step and suitable regularity conditions, suppose $V_n \xrightarrow{p} V$, where V is positive definite. The oracle two-step GMM asymptotic covariance is

$$[G^T V G]^{-1} G^T V \Omega V G [G^T V G]^{-1}.$$

If Ω is positive definite, choosing $V_n \xrightarrow{p} \Omega^{-1}$ reduces the covariance to

$$[G^T \Omega^{-1} G]^{-1}.$$

This is the smallest covariance within the oracle GMM class indexed by the weighting matrix, in the positive semidefinite order.

Theorem 4: the oracle distribution of PDG

For the augmented system, define

$$\Omega^* = \mathbb{E}[\psi^*(\psi^*)^T],$$

where ψ^* denotes $\psi^*(W, \gamma_0, \alpha_0, \theta_0, \mu_0)$.

Theorem 4

Under C1–C8, R1–R7, and S1–S2, suppose Ω^* is positive definite and

$$V_n^* \xrightarrow{P} V^* = (\Omega^*)^{-1}.$$

Then

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, [G^T \Omega^{-1} G]^{-1}\right).$$

PDG matches the optimally weighted oracle debiased GMM distribution.

Interpreting the oracle result

- ▶ The number of moments is fixed. Theorems 3 and 4 are **pointwise results for a fixed DGP**.
- ▶ They do not give uniform approximations when invalid moment means approach zero. PDG is nonregular under support-changing local perturbations of order $n^{-1/2}$ (Liao, 2013; Leeb and Pötscher, 2005).
- ▶ Matching the oracle covariance does not establish **semiparametric efficiency**. The underlying model must specify an efficiency bound, and PDG falls outside the class of regular estimators used to derive such bounds (Ackerberg et al., 2014).

Proposition 1: Adaptive Lasso penalty

$$\hat{P}(\mu_j) = \frac{\lambda_n}{|\tilde{\mu}_j|^r} |\mu_j|, \quad r > 0.$$

Proposition 1

Suppose $\tilde{\mu}$ is a $\sqrt{n}$ -consistent estimator of μ_0 and

$$\sqrt{n}\lambda_n \rightarrow 0, \quad n^{(1+r)/2}\lambda_n \rightarrow \infty.$$

Then the adaptive Lasso penalty satisfies C8, R7, S1, and S2. The scaling sequence in S2 can be $a_n = n^{r/2}$.

- ▶ Adaptive weights remain bounded at nonzero violations and diverge at zero violations.
- ▶ Conditional on the adaptive weights, the *penalty* is convex.

Proposition 2: Initial estimator of μ_0

Use the preliminary estimates already needed for cross-fitting:

$$\tilde{\mu} = \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \left[g_2(W_i, \hat{\gamma}_\ell, \tilde{\theta}_\ell) + \phi_2(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell) \right].$$

Proposition 2

Under C3, C6, and R2–R6, suppose $\tilde{\theta}_\ell$ is $\sqrt{n}$ -consistent for every $\ell = 1, \dots, L$. Then $\tilde{\mu}$ is a $\sqrt{n}$ -consistent estimator of μ_0 .

No additional preliminary estimation is required. The FSIF adjustment is essential to the general $\sqrt{n}$ guarantee.

Preliminary moments and covariance

For $i \in I_\ell$, define

$$\hat{\psi}_i := g(W_i, \hat{\gamma}_\ell, \tilde{\theta}_\ell) + \phi(W_i, \hat{\gamma}_\ell, \hat{\alpha}_\ell, \tilde{\theta}_\ell),$$

$$\bar{\psi} := \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i,$$

$$\hat{\Omega}^* := \frac{1}{n} \sum_{i=1}^n (\hat{\psi}_i - \bar{\psi})(\hat{\psi}_i - \bar{\psi})^T.$$

Centering accounts for the nonzero mean of invalid candidate moments. With feasible optimal weighting,

$$V_n^* = (\hat{\Omega}^*)^{-1}.$$

Adaptive Lasso penalty with $r = 2$

With $r = 2$, use the coordinate-specific adaptive Lasso penalty

$$\hat{P}(\mu_j) = 2\hat{c}_j n^{-1} \frac{|\mu_j|}{|\tilde{\mu}_j|^2}, \quad j = 1, \dots, m_2,$$

where $\tilde{\mu}_j$ is the preliminary estimate of $\mu_{0,j}$. Under optimal weighting, the loading is

$$\hat{c}_j = \left([\hat{\Omega}^*]_{m_1+j, m_1+j}^2 [V_n^*]_{m_1+j, m_1+j} \right)^{1/2}.$$

The loading accounts for score variability and scale differences across candidate moments. It makes the penalty invariant to rescaling the moment conditions.

Post-selection estimation and variance

In practice, we recommend $\hat{\theta}_{\text{post}}$: unpenalized debiased GMM using the baseline and selected valid candidate moments, with the same sample splitting, nuisance estimates, and FSIF corrections.

Let $\hat{\Omega}$ be the principal submatrix of $\hat{\Omega}^*$ corresponding to g_1 and $g_{2,\hat{S}^c}$, and define

$$\hat{G} := \frac{1}{n} \sum_{\ell=1}^L \sum_{i \in I_\ell} \frac{\partial}{\partial \theta^T} \begin{bmatrix} g_1(W_i, \hat{\gamma}_\ell, \hat{\theta}) \\ g_{2,\hat{S}^c}(W_i, \hat{\gamma}_\ell, \hat{\theta}) \end{bmatrix}.$$

With optimal weighting, use the variance estimator

$$\frac{1}{n} \left[\hat{G}^T \hat{\Omega}^{-1} \hat{G} \right]^{-1}.$$

Proposition 3: Post-selection PDG

Proposition 3

Under C1–C8, R1–R7, and S1–S2, suppose Ω^* is positive definite and

$$V_n^* \xrightarrow{P} V^* = (\Omega^*)^{-1}.$$

Then

$$\sqrt{n}(\hat{\theta}_{\text{post}} - \theta_0) \xrightarrow{d} N\left(0, [G^T \Omega^{-1} G]^{-1}\right).$$

- ▶ Post-PDG and optimally weighted PDG have the same first-order distribution as the oracle debiased GMM estimator.
- ▶ The result is pointwise for a fixed DGP and a fixed number of moments. Refitting can improve finite-sample bias.

Simulation: partially linear IV model

$U_i = (U_{i,1}, \dots, U_{i,10})$ has independent $\text{Unif}[-1, 1]$ coordinates; only the first five enter the true conditional means.

$$Y_i = X_i \theta_0 + h_0(U_i) + \epsilon_i, \quad \theta_0 = 0.5,$$

$$b(U) = [\sin(\pi U_1), U_2^2 - 1/3, U_3, \cos(\pi U_4), U_5^3]^T,$$

$$r_0(U) = (0.6, 0.5, 0.4, 0.3, 0.2)b(U),$$

$$h_0(U) = (0.7, -0.6, 0.5, 0.4, -0.3)b(U),$$

$$X_i = r_0(U_i) + \pi_1 A_{0i} + (0.15, 0.15, 0.20, 0.20, 0.25)(A_{1i}, \dots, A_{5i})^T \\ + 0.05 \sum_{k=1}^{20} F_{ki} + \nu_i,$$

$$Z_{1i} = m_{1,0}(U_i) + A_{0i}, \quad Z_{2,j,i} = m_{2,j,0}(U_i) + A_{ji}, \quad j = 1, \dots, 5,$$

$$Z_{2,5+k,i} = m_{2,5+k,0}(U_i) + F_{ki} + \delta_k \epsilon_i, \quad k = 1, \dots, 20,$$

$$\delta_k = c_{\text{inv}} + (0.8 - c_{\text{inv}})(k - 1)/19.$$

$$A_i = (A_{0i}, \dots, A_{5i})^T \sim N(0, \Sigma_A), \quad [\Sigma_A]_{jk} = 0.2^{|j-k|}, \quad j, k = 0, \dots, 5;$$

$F_i \sim N(0, I_{20})$. The unit-variance Gaussian errors (ϵ_i, ν_i) have correlation 0.6;

the blocks $U_i, A_i, F_i, (\epsilon_i, \nu_i)$ are independent. The functions $m_{1,0}(U)$ and

$m_{2,j,0}(U)$, $j = 1, \dots, 25$, use the paper's conditional-mean specification.

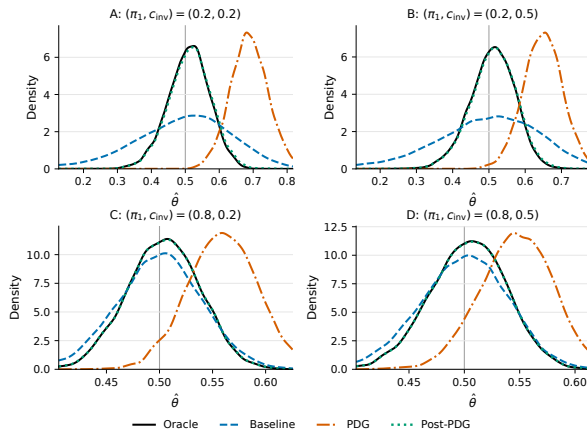
$\ell_0(U) = E[Y | U] = \theta_0 r_0(U) + h_0(U)$. Thus $\mu_0 = (0, 0, 0, 0, 0, \delta_1, \dots, \delta_{20})^T$.

Table 1: PDG moment selection

Design	n	Unsafe	Exact	Conservative
A	500	0.6716	0.2971	0.0313
A	1000	0.0150	0.9295	0.0555
A	2000	0.0000	0.9795	0.0205
B	500	0.0330	0.8528	0.1142
B	1000	0.0001	0.9504	0.0495
B	2000	0.0000	0.9826	0.0174
C	500	0.3124	0.6685	0.0191
C	1000	0.0062	0.9843	0.0095
C	2000	0.0000	0.9973	0.0027
D	500	0.0000	0.9716	0.0284
D	1000	0.0000	0.9885	0.0115
D	2000	0.0000	0.9981	0.0019

Selection probabilities. Unsafe: any invalid candidate retained. Exact: all and only valid candidates retained. Conservative: only a strict subset of valid candidates retained.

Sampling distributions at $n = 1000$



Vertical line: $\theta_0 = 0.5$. Densities use all draws; at most 3.32% of any estimator's draws lie outside its panel.

Bias and RMSE at $n = 1000$

Design	Bias		RMSE			
	PDG	Post-PDG	Oracle	PDG	Post-PDG	Baseline
A	0.1906	0.0158	0.0630	0.1992	0.0669	0.1526
B	0.1504	0.0135	0.0626	0.1603	0.0650	0.1518
C	0.0592	0.0035	0.0348	0.0678	0.0349	0.0407
D	0.0472	0.0037	0.0344	0.0574	0.0345	0.0404

- ▶ Post-PDG's RMSE is close to the Oracle benchmark.
- ▶ Direct PDG retains substantial shrinkage bias despite low dispersion.
- ▶ Across all twelve cells, Post-PDG has lower absolute bias and RMSE than PDG.

Bias and RMSE relative to $\theta_0 = 0.5$. Full tables: A, B, C, D.

Table 7: variance gains, Designs A and B

$\pi_1 = 0.2$; $c_{\text{inv}} = 0.2$ (A) or 0.5 (B).

Design	n	ns_{Baseline}^2	Reduction (%)		Variance / Oracle	
			PDG	Post-PDG	PDG	Post-PDG
A	500	111.2296	90.27	81.33	1.0886	2.0878
A	1000	23.1321	85.52	81.75	0.8745	1.1018
A	2000	20.2067	85.69	82.27	0.8202	1.0159
B	500	67617.3915	99.45	99.47	0.6477	0.6193
B	1000	22.9565	86.57	82.40	0.8104	1.0625
B	2000	20.5168	85.83	82.77	0.8364	1.0173

At $n = 500$, extreme estimates make variance comparisons unstable in B.

s_e^2 : empirical variance. Reduction: $100(1 - s_e^2/s_{\text{Baseline}}^2)$. Variance comparisons do not account for bias.

Table 7: variance gains, Designs C and D

$\pi_1 = 0.8$; $c_{\text{inv}} = 0.2$ (C) or 0.5 (D).

Design	n	ns_{Baseline}^2	Reduction (%)		Variance / Oracle	
			PDG	Post-PDG	PDG	Post-PDG
C	500	1.8719	31.34	21.25	0.9478	1.0871
C	1000	1.6575	34.40	27.43	0.9078	1.0044
C	2000	1.5626	31.40	26.89	0.9392	1.0009
D	500	1.8730	33.39	26.18	0.9063	1.0045
D	1000	1.6283	34.90	27.68	0.9042	1.0045
D	2000	1.5563	30.59	26.33	0.9437	1.0016

At $n = 1000, 2000$, Post-PDG reductions closely match the population Oracle gains: 81.71% for $\pi_1 = 0.2$ and 26.49% for $\pi_1 = 0.8$.

s_e^2 : empirical variance. Reduction: $100(1 - s_e^2/s_{\text{Baseline}}^2)$. Variance comparisons do not account for bias.

Simulation interpretation and limitations

- ▶ Valid additional moments produce substantial precision gains. At $n = 1000, 2000$, Post-PDG combines these gains with small bias, near-Oracle RMSE, and coverage of 92.34–94.81%.
- ▶ Direct PDG's positive loadings continue to penalize nonzero auxiliary coefficients, producing substantial shrinkage bias.
- ▶ In Liao (2013), loadings for moments classified as invalid in a preliminary shrinkage fit become zero. Our loadings are strictly positive and do not depend on a preliminary support estimate.
- ▶ At $n = 500$, selection errors and undercoverage remain important; extreme estimates in Design B destabilize variance comparisons.
- ▶ **Recommended final estimator: Post-PDG.** Unpenalized refitting removes continuing penalization, but does not make finite-sample moment-selection uncertainty disappear.

Conclusion

- ▶ Orthogonalization and cross-fitting preserve the oracle properties of penalized moment selection with estimated first steps.
- ▶ Under the stated conditions, PDG recovers the invalid set and attains the oracle debiased GMM distribution for a fixed DGP.
- ▶ Post-PDG reduces direct shrinkage bias and captures substantial precision gains in the simulations.
- ▶ Future work: uniform post-selection inference, a growing number of moments, and characterization of semiparametric efficiency.

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Table 2: Design A, $n = 500$

$$(\pi_1, c_{\text{inv}}) = (0.2, 0.2).$$

Estimator	Bias	SD	RMSE
Oracle	0.0264	0.1410	0.1435
PDG	0.4316	0.1471	0.4560
Post-PDG	0.1480	0.2038	0.2518
Baseline	-0.0259	0.4717	0.4723
NPG	0.4346	0.1119	0.4488

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical.

Results at $n = 1000, 2000$

Main comparison

Table 2: Design A, $n = 1000, 2000$

$$(\pi_1, c_{\text{inv}}) = (0.2, 0.2).$$

Estimator	$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0119	0.0619	0.0630	0.0054	0.0420	0.0423
PDG	0.1906	0.0579	0.1992	0.0883	0.0380	0.0962
Post-PDG	0.0158	0.0650	0.0669	0.0060	0.0423	0.0427
Baseline	-0.0123	0.1521	0.1526	-0.0042	0.1005	0.1006
NPG	0.2170	0.0578	0.2245	0.1055	0.0373	0.1119

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical.

Results at $n = 500$

Main comparison

Table 3: Design B, $n = 500$

$$(\pi_1, c_{\text{inv}}) = (0.2, 0.5).$$

Estimator	Bias	SD	RMSE
Oracle	0.0261	1.0712	1.0715
PDG	0.3656	0.8621	0.9364
Post-PDG	0.0602	0.8430	0.8451
Baseline	-0.1598	11.6290	11.6296
NPG	0.4063	0.1166	0.4227

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical. Extreme estimates make dispersion comparisons unstable.

Results at $n = 1000, 2000$

Main comparison

Table 3: Design B, $n = 1000, 2000$

$$(\pi_1, c_{\text{inv}}) = (0.2, 0.5).$$

Estimator	$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0109	0.0617	0.0626	0.0049	0.0417	0.0420
PDG	0.1504	0.0555	0.1603	0.0701	0.0381	0.0798
Post-PDG	0.0135	0.0636	0.0650	0.0054	0.0420	0.0424
Baseline	-0.0091	0.1515	0.1518	-0.0057	0.1013	0.1014
NPG	0.1885	0.0536	0.1960	0.0922	0.0369	0.0993

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical.

Results at $n = 500$

Main comparison

Table 4: Design C, $n = 500$

$$(\pi_1, c_{\text{inv}}) = (0.8, 0.2).$$

Estimator	Bias	SD	RMSE
Oracle	0.0080	0.0521	0.0527
PDG	0.1298	0.0507	0.1393
Post-PDG	0.0160	0.0543	0.0566
Baseline	-0.0012	0.0612	0.0612
NPG	0.1186	0.0551	0.1308

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical.

Results at $n = 1000, 2000$

Main comparison

Table 4: Design C, $n = 1000, 2000$

$$(\pi_1, c_{\text{inv}}) = (0.8, 0.2).$$

Estimator	$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0034	0.0346	0.0348	0.0013	0.0239	0.0239
PDG	0.0592	0.0330	0.0678	0.0279	0.0232	0.0363
Post-PDG	0.0035	0.0347	0.0349	0.0013	0.0239	0.0239
Baseline	-0.0006	0.0407	0.0407	-0.0006	0.0280	0.0280
NPG	0.0579	0.0338	0.0671	0.0288	0.0232	0.0370

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical.

Results at $n = 500$

Main comparison

Table 5: Design D, $n = 500$

$$(\pi_1, c_{\text{inv}}) = (0.8, 0.5).$$

Estimator	Bias	SD	RMSE
Oracle	0.0080	0.0525	0.0531
PDG	0.1036	0.0500	0.1150
Post-PDG	0.0080	0.0526	0.0532
Baseline	-0.0018	0.0612	0.0612
NPG	0.1147	0.0520	0.1259

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical.

Results at $n = 1000, 2000$

Main comparison

Table 5: Design D, $n = 1000, 2000$

$$(\pi_1, c_{\text{inv}}) = (0.8, 0.5).$$

Estimator	$n = 1000$			$n = 2000$		
	Bias	SD	RMSE	Bias	SD	RMSE
Oracle	0.0037	0.0342	0.0344	0.0018	0.0239	0.0240
PDG	0.0472	0.0326	0.0574	0.0227	0.0232	0.0325
Post-PDG	0.0037	0.0343	0.0345	0.0018	0.0239	0.0240
Baseline	-0.0005	0.0404	0.0404	-0.0002	0.0279	0.0279
NPG	0.0547	0.0328	0.0638	0.0275	0.0232	0.0360

Bias and RMSE are relative to $\theta_0 = 0.5$; SD is empirical.

Results at $n = 500$

Main comparison

Table 6: standard errors and coverage, A

$(\pi_1, c_{\text{inv}}) = (0.2, 0.2)$; nominal 95% confidence intervals.

Estimator	$n = 500$		$n = 1000$		$n = 2000$	
	Avg. SE	Coverage	Avg. SE	Coverage	Avg. SE	Coverage
Oracle	0.1022	0.9243	0.0617	0.9342	0.0419	0.9433
PDG	0.0953	0.0076	0.0619	0.1215	0.0419	0.4335
Post-PDG	0.0953	0.6350	0.0619	0.9234	0.0419	0.9406
Baseline	0.2745	0.9517	0.1493	0.9534	0.0995	0.9516
NPG	0.0941	0.0074	0.0687	0.0870	0.0481	0.3870

Avg. SE: mean estimated standard error. Coverage: unconditional probability for intervals centered at each estimator. NPG omits nuisance correction.

Main coverage comparison

Table 6: standard errors and coverage, B

$(\pi_1, c_{\text{inv}}) = (0.2, 0.5)$; nominal 95% confidence intervals.

Estimator	$n = 500$		$n = 1000$		$n = 2000$	
	Avg. SE	Coverage	Avg. SE	Coverage	Avg. SE	Coverage
Oracle	0.1240	0.9314	0.0616	0.9371	0.0419	0.9459
PDG	0.1142	0.0446	0.0618	0.3026	0.0420	0.6093
Post-PDG	0.1142	0.8911	0.0618	0.9295	0.0420	0.9436
Baseline	0.4487	0.9568	0.1490	0.9520	0.0996	0.9514
NPG	0.0994	0.0127	0.0686	0.1683	0.0482	0.5169

Avg. SE: mean estimated standard error. Coverage: unconditional probability for intervals centered at each estimator. NPG omits nuisance correction.

Main coverage comparison

Table 6: standard errors and coverage, C

$(\pi_1, c_{\text{inv}}) = (0.8, 0.2)$; nominal 95% confidence intervals.

Estimator	$n = 500$		$n = 1000$		$n = 2000$	
	Avg. SE	Coverage	Avg. SE	Coverage	Avg. SE	Coverage
Oracle	0.0521	0.9404	0.0345	0.9439	0.0238	0.9477
PDG	0.0519	0.2997	0.0345	0.5895	0.0238	0.7766
Post-PDG	0.0519	0.9196	0.0345	0.9430	0.0238	0.9477
Baseline	0.0621	0.9544	0.0407	0.9487	0.0279	0.9500
NPG	0.0551	0.4209	0.0386	0.6837	0.0271	0.8381

Avg. SE: mean estimated standard error. Coverage: unconditional probability for intervals centered at each estimator. NPG omits nuisance correction.

Main coverage comparison

Table 6: standard errors and coverage, D

$(\pi_1, c_{\text{inv}}) = (0.8, 0.5)$; nominal 95% confidence intervals.

Estimator	$n = 500$		$n = 1000$		$n = 2000$	
	Avg. SE	Coverage	Avg. SE	Coverage	Avg. SE	Coverage
Oracle	0.0521	0.9391	0.0345	0.9468	0.0238	0.9482
PDG	0.0521	0.4862	0.0345	0.7155	0.0238	0.8319
Post-PDG	0.0521	0.9388	0.0345	0.9461	0.0238	0.9481
Baseline	0.0621	0.9514	0.0407	0.9501	0.0279	0.9493
NPG	0.0554	0.4522	0.0385	0.7220	0.0271	0.8496

Avg. SE: mean estimated standard error. Coverage: unconditional probability for intervals centered at each estimator. NPG omits nuisance correction.

Main coverage comparison